

TorchJD: Jacobian Descent in PyTorch

Valérian Rey — valerian.rey@gmail.com
Pierre Quinton — pierre.quinton@epfl.ch



github.com/SimplexLab/TorchJD



torchjd.org

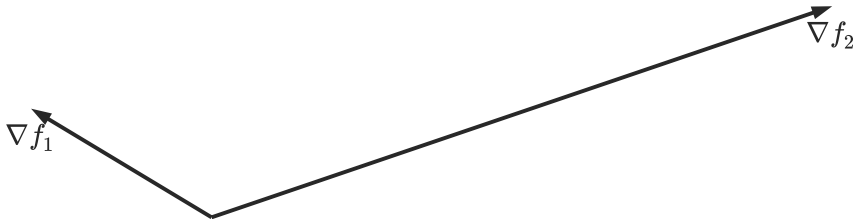


arxiv.org/pdf/2406.16232

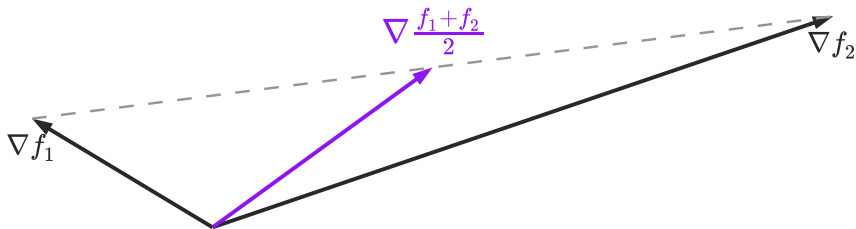
Motivation

Jacobian descent

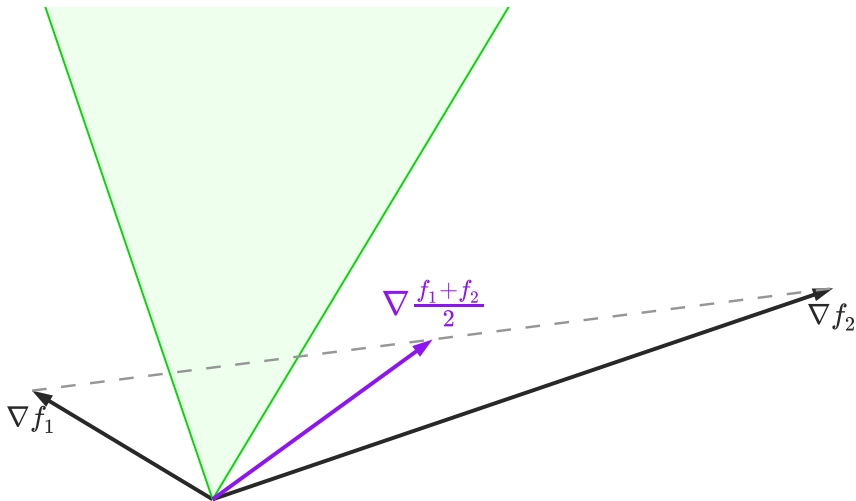
Jacobian descent



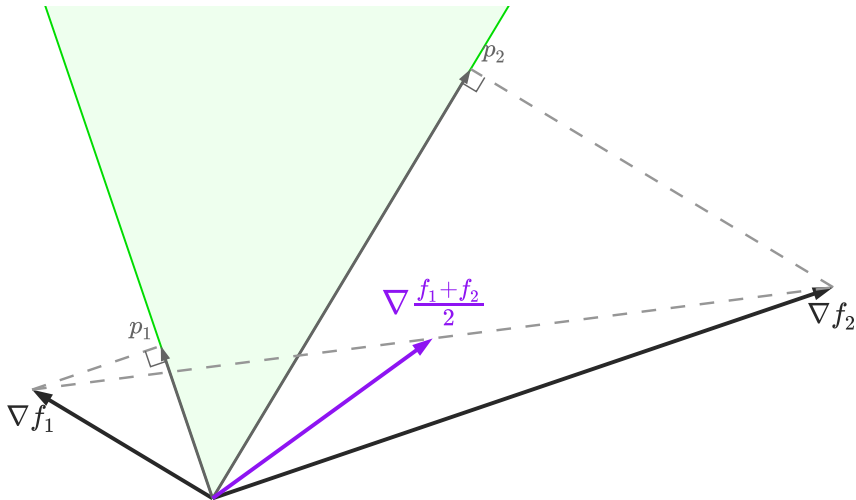
Jacobian descent



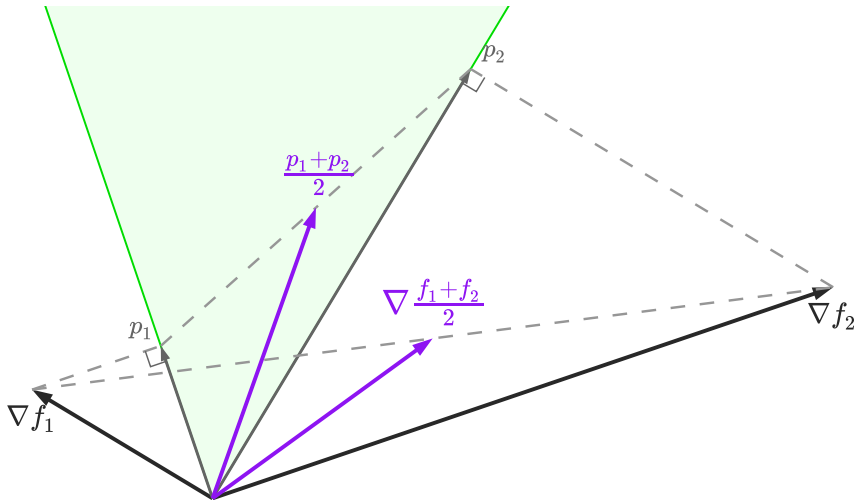
Jacobian descent



Jacobian descent



Jacobian descent



TorchJD

TorchJD: core functions

Library	Function	Description
torch	autograd.grad	compute gradients
torch	autograd.backward	compute gradients and accumulate in .grad

TorchJD: core functions

Library	Function	Description
torch	autograd.grad	compute gradients
torch	autograd.backward	compute gradients and accumulate in .grad
torchjd	autojac.jac	compute jacobians

TorchJD: core functions

Library	Function	Description
torch	autograd.grad	compute gradients
torch	autograd.backward	compute gradients and accumulate in .grad
torchjd	autojac.jac	compute jacobians
torchjd	autojac.backward	compute jacobians and accumulate in .jac

TorchJD: core functions

Library	Function	Description
torch	autograd.grad	compute gradients
torch	autograd.backward	compute gradients and accumulate in .grad
torchjd	autojac.jac	compute jacobians
torchjd	autojac.backward	compute jacobians and accumulate in .jac
torchjd	autojac.jac_to_grad	aggregate .jac fields into .grad

TorchJD: basic usage

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
loss_fn = MSELoss(reduction="none")

for x, y in dataloader:
    z = model(x) # shape: [16]
    losses = loss_fn(z, y) # shape: [16]
    autojac.backward(losses)
    # parameters now have a .jac field
    autojac.jac_to_grad(model.parameters(), UPGrad())
    # parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

More examples at torchjd.org.

Discussion